ASL Levelling Port: Build Readiness and Build Guide

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Verdict

Once it is wired in, the port should compile and link in the Keil build. As delivered, it will not run: nothing includes its make fragments or calls the adapter, so the linker drops the whole chain.

Question	Answer	Basis
Compiles with the ILCU flags?	Very likely	0 errors on host with the documented Keil flags and the App defines; 20 warnings, all in unchanged DPE lines
Links?	Very likely, once the sources are listed	Needs only 5 IoHwAb functions that exist in ILCU, plus 3 armclang runtime helpers
Ends up in the hex?	No, until `Rte.c` calls the adapter	`armlink --lto --remove` strips unreferenced code (`BuildSystem.md:293`)
Runs safely on the board?	Not proven	512-byte task stack, output-stage scaling (F3), VBAT channel (F7), uninitialised locals in CtrlDcm

Every Keil-side answer is UNVERIFIED: this workspace has no armclang, no TI MCAL and no `Build/` tree.

- **What could still break the build** is exactly what is missing here: the `Build/` tree, the real TI headers (`Std_Types.h`, `Compiler.h` and `Platform_Types.h` ran as stand-ins) and armclang 6.16 itself.
- **I was wrong about the project folder.** PORT_CHANGES.md §2 and §8 copy the platform's `Al077`. ILCU's configuration is `Al077a`, and the platform lists an `Al077a H00 Proto App` target (`ecu9_core_ptf/.vscode/settings.json:6`). Use `Cust/Al/Al077a/App` and `Al077a` in `Build.bat`; UNVERIFIED until you open the Build tree.

- **Start with** improvements 1 to 4, then follow the Build guide.

Scope and method

The analysis covers the 35 ported files, the 4 make fragments and every ILCU file they touch, checked against the platform's build guide (`src_AL/ecu9_core_ptf/tools/docs/BuildSystem.md`). Six host checks ran on this machine; four things can only be checked in the real build.

Host check	Question it answers	Result
Compile for `thumbv6m` with `-std=gnu17 -fshort-enums -Os -fno-inline-functions -ffunction-sections -fdata-sections` and the App defines	Does it compile with the documented Keil flags (`BuildSystem.md:271`)?	0 errors; the same 20 DPE warnings with `AUTOSAR_431` or `AUTOSAR_440`
Object build and symbol tables, 13 objects	What must the rest of ILCU provide at link time?	5 IoHwAb functions plus `__aeabi_idiv`, `__aeabi_uidiv`, `__aeabi_memcpy`; no duplicate definitions
`-fstack-usage` plus a call graph from the assembly	How deep does each runnable go?	128 B worst case on the 5 ms path, 88 B for Init (ported code only)
Name scan: every global, macro, type and enumerator of the port against all non-ported ILCU sources	Does any name clash?	3 known overlaps, all safe
Include and file-list scan, exact case	Does every path resolve on a case-sensitive file system?	All resolve; source basenames unique
`-Weverything`	What would a stricter warning set add?	462 warnings in DPE files, 3 on adapter types, 0 errors

Behaviour was verified earlier by the host harness: 4 scenarios at `-O0` and `-O2` (see the port report).

Not checkable here:

- The `Build/` tree: the CustVar files, the real `-W` list in `ToolChain_030_Flags.gmk`, the values of `PATH_SRC_CDD_BASE` and `PATH_SRC_BSW`.
- `Src_ThirdParty/`: the TI MCAL 00.09.00.00 headers and the FuSa-RTS sources.
- armclang 6.16 itself; the host used Apple clang 21.
- The board: stack watermark, PWM and ADC scaling.

Findings by area

Build wiring is the only build blocker; headers, symbols, names and memory are clean. The real risks are at run time: stack depth, hardware scaling and undefined behaviour inherited from DPE.

Build time

Area	Finding	Evidence	Risk
Make fragments	All 21 entries resolve with exact case, but nothing includes them yet. The `C_INCLUDES := … $(C_INCLUDES)` prepend is the documented convention.	`BuildSystem.md:281`	Blocker until applied
Project name	ILCU uses the `Al077a` configuration, not `Al077` as PORT_CHANGES.md says.	`config_AL/Al/Al077a`, `.vscode/settings.json:6`	Lines in the wrong folder are never read
Path variables	`PATH_SRC_CDD_BASE` comes from the ILCU Build tree (the CddAsm330lhh fragments use it). `PATH_SRC_BSW` must point at `ecu9_core_ptf/Src/Src_AL/bsw`.	`CddAsm330lhh_030_Flags.gmk:5`, `BuildSystem.md:192`	`IoHwAb_Internals.h` not found if wrong
Scheduling	Nothing calls the adapter yet, and the linker removes unreferenced sections.	`Rte.c:98-106`, `Rte.c:156-188`, `BuildSystem.md:293`	Chain absent from the hex
External headers	Only `Std_Types.h` (TI), `stdbool.h` and `stdlib.h` (armclang) come from outside the snapshot. The port uses `STATIC`, `INLINE` (30×), `NULL_PTR`, `uint8_least` and the usual types. ILCU itself writes `STATIC INLINE`, so `INLINE` is non-static, as the port assumes.	`Os_Wrapper.c:134`	Low

Area	Finding	Evidence	Risk
Link closure	`IoHwAb_AnalogReadConverted`, `IoHwAb_TimingSetDutyCycle`, `IoHwAb_TimingStopActuation`, `IoHwAb_DiscreteEN_LWR_Write`, `IoHwAb_VoltageKL30_Read`, plus 3 runtime helpers. No float, no 64-bit arithmetic.	`IoHwAb_Analog.c:107`, `IoHwAb_Timing.c:78`, `:113`, `IoHwAb_Cfg.c:45`, `:58`	Low
Name overlaps	`DEM_EVENT_STATUS_PASSED/FAILED` (guarded) and `NvM_BlockIdType` (identical typedef, legal in gnu17). Third-party code was not scanned.	`ILCU_LevellingAdapter.h:96-101`, `:531`	Low
Warnings	`-Wall` gives 20 (DPE lines only), `-Wextra` adds 1, `-Weverything` gives 462 in DPE files (317 are `-Wdocumentation`). ILCU's own `swc_imu` and `swc_sbc_demo` give 1 under `-Weverything`. No `-Werror` is documented.	`BuildSystem.md:262`, `:272`	Noise, unless `-Werror` is set
Unit-test include	`#include <RteCommonTypes.h>` is active only when `UNIT_TEST` is defined.	`CtrlDcm_Data.h:27-29`	Test builds only
Memory	10.5 KB code, 190 B constants, 322 B RAM, against 512 KB flash and 62 KB RAM.	`BuildSystem.md:290-291`	None

Run time

Area	Finding	Evidence	Risk
Task stack	`BasicTask` has 512 B. The chain needs 128 B, plus the IoHwAb and MCAL leaf calls (not measurable here), plus 64 B of exception and context frame. An overflow ends in the RTX error hook's endless loop.	`Os_Wrapper_Cfg.c:124`, `RTX_Config.h:207`, `RTX_Config.c:54`	Medium, unmeasured

Area	Finding	Evidence	Risk
Init order	IoHwAb, Pwm and Adc are initialised before the OS starts. The 5 ms alarm is armed before `Rte_Start`, but `BasicTask` cannot run first unless `InitTask` blocks (same priority, no round-robin).	`EcuM_Callout_Stubs.c:188-194`, `EcuM.c:106`, `:115`, `RTX_Config.h:49`	Low
Hardware ownership	The adapter is the only user of CTRL_LWR (TIMG9 C1, PA2), EN_LWR and ADIN_3 (PA15).	`IoHwAb_TimingCfg.c:41-46`, `IoHwAb_AnalogCfg.c:92-100`	None
Output stage (F3)	CddDcm regulates ‰ of KL15 for the DPE output stage. On the ILCU stage, 430 ‰ commanded reaches 225 ‰ at full PWM, and no error is raised.	PORT_CHANGES.md §6.1	High with a real actuator
Analog config (F7)	VBAT is read from ADC0 ch4 (schematic: ch5); the LWR divider is 82.5k/21k (schematic: 82.5k/15k). VBAT outside 8.59–17.69 V locks the output.	`IoHwAb_AnalogCfg.c:251-260`	High with a real actuator

Area	Finding	Evidence	Risk
DPE undefined behaviour	Uninitialised locals on the CtrlDcm recovery path; the harness output already differs between `-O0` and `-O2`.	`CtrlDcmSm.c:902`, `:1293`, `CtrlDcm_RteAccess.c:75`, `:114`, `:151`	Medium

Improvements

Four changes make the first build meaningful, five more belong before the board drives anything, and the rest can wait. Items marked "Claude, on approval" touch only port files; the others touch existing ILCU files or the Build tree, which the brief kept off limits to me.

Before the first build

#	Improvement	Why · evidence	Owner · effort	Status
1	Add the CustVar lines for both fragment pairs (Build guide steps 2 and 3).	Without them nothing is compiled.	You, Build tree · 10 min	Open
2	Confirm `PATH_SRC_CDD_BASE` and `PATH_SRC_BSW` in `_compile_paths.txt`.	`:=` expands once; a wrong value gives dead include paths with no error at the make level.	You · 5 min	Open
3	Add the two `Rte.c` hooks: Init as the first line of `Rte_Start`, MainFunction right after `IoHwAb_MainFunction`.	Otherwise the linker drops the chain (`BuildSystem.md:293`).	You, existing ILCU file · 5 min	Open
4	Correct `Al077` to `Al077a` in PORT_CHANGES.md §2 and §8.	My error: lines written into the wrong Cust folder are never read.	Claude, on approval · 2 min	Open

Before the bench test

#	Improvement	Why · evidence	Owner · effort	Status
5	Measure the `BasicTask` stack; raise `Os_Task_BasicTask_stk[64]` to `[96]` if less than 128 B stays free.	512 B budget; the chain alone needs 128 B before IoHwAb, MCAL and 64 B of frames (`Os_Wrapper_Cfg.c:124`).	You, ILCU OS config · 30 min	Open
6	Initialise the 5 uninitialised CtrlDcm locals to their safe values, marked `ILCU_PORT`.	Undefined behaviour: `-O0` and `-O2` already disagree, so Keil `-Os` with LTO is unknown. Reverses decision Q3 for these lines only.	Claude, on approval · 15 min plus harness re-run	Open
7	Add an init guard: MainFunction returns until Init has run.	Removes the reliance on `InitTask` never blocking before `Rte_Start` (`EcuM.c:106-115`).	Claude, on approval · 5 min	Open
8	Decide F3: scale CddDcm's target to the ILCU output stage, or change the hardware.	The lamp silently ends at the wrong position.	You, product and hardware · open	Open
9	Fix F7 in the Al077a IoHwAb analog config (VBAT channel, LWR divider).	VBAT outside 8.59–17.69 V locks CddDcm's output permanently.	You, existing ILCU config · 30 min	Open

Later

#	Improvement	Why · evidence	Owner · effort	Status
10	Add public IoHwAb wrappers for CTRL_LWR and the LWR feedback; then drop `IoHwAb_Internals.h` and its include path.	The adapter reaches into an internal header (`ILCU_LevellingAdapter.c:19`).	You, IoHwAb config · 30 min	Open

#	Improvement	Why · evidence	Owner · effort	Status
11	Remove the 20 DPE warnings by deleting dead DPE diagnostic code. Avoid `FILES_NOWARN_COMPILER`: it would also hide the 5 uninitialised-variable warnings.	Clean logs make new warnings visible (`BuildSystem.md:263`). Reverses Q3.	Claude, on approval · 30 min	Open
12	Plug in the IMU: implement `ILCU_LevellingAdapter_GetPitchAngle` in `swc_imu` and add `ILCU_LVLADP_EXTERNAL_PITCH_SOURCE` to `C_DEFINES`. Convert cdeg to 0.1 mrad as `(sint32)Pitch_cdeg * 349 / 200`, check the sign, return `E_NOT_OK` until `Rte_ImuInitResult == E_OK`.	The stub holds the pitch at 0 (`ILCU_LevellingAdapter.c:119-126`).	IMU team · 1 h	Open
13	Make `ILCU_LVLADP_STUB_PITCH_0P1MRAD` overridable with `#ifndef`, so bench tests can set it from `C_DEFINES`.	Today a test means editing the header (`ILCU_LevellingAdapter.h:45`).	Claude, on approval · 2 min	Open
14	Real NvM blocks and coding (stubs S9 and S10).	LWR position and headlight angle are lost at every reset.	You, ILCU NvM config · 1–2 days	Open
15	Before any host test build with `UNIT_TEST`, remove or provide `RteCommonTypes.h`.	`CtrlDcm_Data.h:27-29`	Claude, on approval · 5 min	Open

Build guide

Seven steps take an unchanged tree to a verified hex. Build once without the port first, so every difference afterwards belongs to the port.

Before you start: a Windows machine with ToolsGeneric (Keil 6.16 armclang and GNU make, `BuildSystem.md` §2), and the ILCU repository with its `Build/` and `Src_ThirdParty/` folders.

1. Baseline build

```
cd <ILCU>\Build\VS
..\Build.bat H00 MSPM0G_3519 ARM_KEIL CUnit GCOV M0P Al077a App default
NonKey rebuild *> $env:TEMP\build_base.log
```

- `Al077a` is UNVERIFIED (derived from `BuildSystem.md:389` and `.vscode/settings.json:6`); use your team's ILCU target if it differs.
- `Build.bat` always exits 0. Success means no `error` in the log and a fresh hex under `Build\Bin\<NAME>\` (`BuildSystem.md:19-24`).
- Keep the warning count and `<NAME>_Size.txt` for comparison.

2. Add the include paths

In `Build\Make\Cust\Al\Al077a\App\CustVar_030_Flags.gmk`, next to the CddAsm330lhh wiring:

```
include
$(PATH_SRC_CDD_BASE)/../app/swc_levelling/Make/IlcuLevelling_030_Flags.gmk
include $(PATH_SRC_CDD_BASE)/CddDcm/Make/CddDcm_030_Flags.gmk
C_INCLUDES := $(C_INCLUDES_ILCULEVELLING) $(C_INCLUDES_CDDDCM) $(C_INCLUDES)
```

- If the file includes the CddAsm330lhh fragment in another form, copy that form.
- The lines belong in the 030 stage: `Make_030_Flags.gmk` resets `C_INCLUDES` before the layers run (`BuildSystem.md:141`).

3. Add the sources

In `CustVar_050_Files.gmk`:

```
include
$(PATH_SRC_CDD_BASE)/../app/swc_levelling/Make/IlcuLevelling_050_Files.gmk
include $(PATH_SRC_CDD_BASE)/CddDcm/Make/CddDcm_050_Files.gmk
FILES_SRC_CUST += $(FILES_SRC_ILCULEVELLING)
FILES_SRC_CDD  += $(FILES_SRC_CDDDCM)
```

- Wire both pairs together: CddDcm includes the adapter header, and CtrlDcm and the adapter include `CddDcm.h`.

4. Compile-only build

Run `rebuild`, since new files are not picked up reliably by `all` (`BuildSystem.md:320`). Then check `Build\Tmp\<NAME>\obj\`:

- `_compile_paths.txt` lists `src_AL/app/swc_levelling` with its `AvacStat1`, `CtrlDcm` and `CtrlDcm/Api` folders, `src_AL/cdd/CddDcm/Api`, `Gen` and `Cfg`, and `…/bsw/IoHwAb/internal`. No entry starts with `/../`.
- `_object_files.txt` lists the 13 new objects: `ILCU_LevellingAdapter`, `AvacStat1`, `AvacStat1Data`, `AvacStat1Init`, `AvacStat1Util`, `ctaaAvacStat1`, `CtrlDcm`, `CtrlDcmSm`, `CtrlDcm_Data`, `CtrlDcm_RteAccess`, `ctaaCtrlDcm`, `CddDcmCtrl`, `CddDcmHal`.
- `err\ILCU_LevellingAdapter.err` is empty; the other new `.err` files hold only the 20 warnings listed in PORT_CHANGES.md §6.4.

- The map shows the chain as removed. That is expected without hooks.

5. Add the hooks

In `src_AL/rte/Rte.c`:

```
#include "ILCU_LevellingAdapter.h"        /* with the other includes,
Rte.c:25-38 */

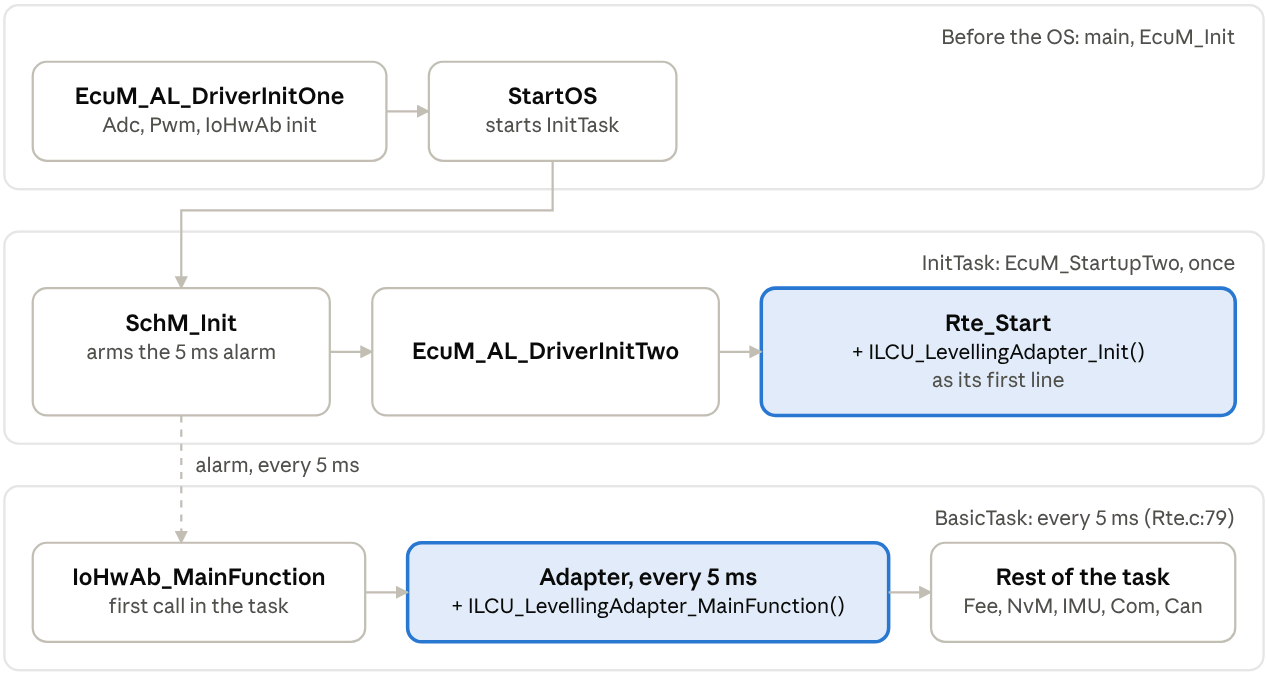
FUNC(Std_ReturnType, RTE_CODE) Rte_Start(void)
{
  Std_ReturnType RetVal;

  ILCU_LevellingAdapter_Init();           /* first: IoHwAb is up, nothing has
blocked yet */
  swc_imu_init();
  RetVal = CddAsm330lhh_Init(&CddAsm330lhh_Config_Default);
  Rte_ImuInitResult = E_NOT_OK;
  return RetVal;
}

TASK(BasicTask)
{
  IoHwAb_MainFunction();
  ILCU_LevellingAdapter_MainFunction();   /* every 5 ms, after IoHwAb
refreshes its inputs */
  Fee_MainFunction();
  /* ... rest unchanged ... */
}
```

The two hooks run after IoHwAb: Init once, MainFunction every 5 ms

Accent: the two calls you add to Rte.c (Build guide step 5)



startup and 5 ms task order · 2 hooks added

Init sits at the top of `Rte_Start`, after EcuM has initialised IoHwAb; MainFunction follows `IoHwAb_MainFunction` in the 5 ms task (`EcuM_Callout_Stubs.c:194`, `EcuM.c:106-115`, `Rte.c:158`).

6. Rebuild and check the link

- `<NAME>.map` contains `ILCU_LevellingAdapter_MainFunction`, `rpAvacStat1_Cycle0`, `rpCtrlDcm_Cycle` and `rpCddDcm_Cycle0`.
- `<NAME>_Size.txt` shows about +11 KB flash and +0.3 KB RAM against the baseline (host estimate, ±20 %).

7. Check the stack before connecting anything

- Static: add `stack` to the existing armlink `--info=` list (`BuildSystem.md:293`), or add `--callgraph`, and read the maximum depth reported for `Os_Task_BasicTask`. Confirm both option names with `armlink --help` for 6.16.
- Dynamic: set `OS_STACK_WATERMARK` to 1 (`RTX_Config.h:214`), run for 1 minute, and read `osThreadGetStackSpace()` for `BasicTask` in the debugger.
- Keep at least 128 B free; otherwise apply improvement 5.

Troubleshooting

Most failures come from one of three causes: a fragment that is not included, a path variable with a different value, or a real TI header that differs from the stand-in. Make runs with `-k`, so the first real error can sit far above the end of the log (`BuildSystem.md:379`).

Symptom	Likely cause	Fix
`'ILCU_LevellingAdapter.h' file not found` in a CddDcm file	IlcuLevelling 030 fragment or the `C_INCLUDES` line missing	Step 2, all three lines
`'CddDcm.h' file not found` in CtrlDcm or the adapter	CddDcm 030 fragment missing	Step 2
`'IoHwAb_Internals.h' file not found`	`PATH_SRC_BSW` is not `…/ecu9_core_ptf/Src/Src_AL/bsw`	Correct the path in `IlcuLevelling_030_Flags.gmk:18`, then check `_compile_paths.txt`
Entries starting with `/../app/` in `_compile_paths.txt`	`PATH_SRC_CDD_BASE` was empty when `:=` expanded	Set `PATH_SRC_ILCULEVELLING` before the include (PORT_CHANGES.md §2)
`L6218E` undefined `riCddDcm_Init` or `rpCddDcm_Cycle0`	`FILES_SRC_CDDDCM` not appended	Step 3, `FILES_SRC_CDD +=`
`L6218E` undefined `rpCtrlDcm_Cycle`, `riAvacStat1_Init` and others	`FILES_SRC_ILCULEVELLING` not appended	Step 3, `FILES_SRC_CUST +=`
`L6218E` undefined `IoHwAb_Timing…` or `IoHwAb_AnalogReadConverted`	IoHwAb internal sources not built in this configuration	Check that `Bsw_050_Files.gmk` lists `IoHwAb_Timing` and `IoHwAb_Analog`
`L6218E` undefined `ILCU_LevellingAdapter_GetPitchAngle`	`ILCU_LVLADP_EXTERNAL_PITCH_SOURCE` is defined, but the IMU side is not written	Remove the define, or implement the function (improvement 12)
`L6200E` symbol multiply defined, e.g. `TimerGetTick`, `NM_GetMode` or an `Rte_*` port	A TI or RTX file outside this snapshot uses the same name; with LTO this shows only at link (`BuildSystem.md:380`)	Rename it in the adapter and its header, marked `ILCU_PORT`
Typedef or macro redefinition in `ILCU_LevellingAdapter.c`	A real TI header pulled in by `IoHwAb.h` defines the same name	Guard the name in the adapter header, as is done for the `DEM_*` macros
`duplicate 'static'` in `AvacStat1Data.h`	The real `Compiler.h` defines `INLINE` as `static inline`	Replace `static INLINE` with `INLINE` there; unlikely, since ILCU writes `STATIC INLINE` itself
`'RteCommonTypes.h' file not found`	`UNIT_TEST` is defined	Do not define it for target builds; improvement 15
Build passes, but no levelling code in the map	No hooks in `Rte.c`	Step 5
More than 20 warnings in the ported files	The real `-W` list is stricter than `-Wall`	Compare with the `-Weverything` classes in Findings; warnings fail the build only with `-Werror`
Board hangs, debugger stops in `osRtxErrorNotify` with `osRtxErrorStackOverflow`	`BasicTask` stack overflow	Improvement 5

On-target checks

Run these with the LWR actuator disconnected, or on a dummy load. Connect the actuator only after F3 and F7 are settled (improvements 8 and 9).

#	Check	How	Pass when
1	Chain runs	Watch `ILCU_LvlAdp_TimeUs`	Grows by 5000 every 5 ms
2	Phase reaches Adj	Watch `ILCU_LvlAdp_CtrlDcmStatus.eASL_PHASE` and `.eASL_MODE`	`eDcmPhase_Adj` (1) within 20 ms, mode `eDcmModeAvailable`, and it stays there
3	Target follows pitch	Set `ILCU_LVLADP_STUB_PITCH_0P1MRAD` to 200 (`ILCU_LevellingAdapter.h:45`), rebuild, watch `ILCU_LvlAdp_DcmTarget.sDcmTarget.nPosition`	−115 (0.01°) after the power-on adjustment, as in the host harness
4	Outputs move	Scope PA3 (EN_LWR) and PA2 (CTRL_LWR) during a target change	EN_LWR high and PWM active while moving, both idle afterwards
5	Supply reading	Compare the value returned by `Rte_Read_IoHwAb_User_GetAdc_DIAG_VBAT` with the bench supply	Inside 8,590–17,690 mV and close to the supply; otherwise F7
6	No electrical errors	Watch `ILCU_LvlAdp_DemStatus[23]` , `[25]` and `ILCU_LvlAdp_DcmError`	Never `0x03` (FAILED); both error flags FALSE
7	Stack headroom	Watermark on, `osThreadGetStackSpace()` for `BasicTask` after 1 minute	At least 128 B free
8	Fail path	Make the pitch stub return `E_NOT_OK`	Phase Fail, target −32 (the coded safety position)

Revert the test edits from checks 3 and 8 afterwards.